Wilhelm Conrad Röntgen: Faustian Genius

By: Peter Polner



Life

- Born 27 March 1845 (Lennep, Prussia)
- 1848 moved to Netherlands (Apeldoorn)
  - Causing him to become stateless
- Wasn’t able to graduate technical school
- Moved between Switzerland and Germany to become a Professor and pursue scientific studies
- 1872 Married Anna Bertha Ludwig (from zürich)
- 1887 Adopted child of Röntgen’s brother
- 1910 Röntgen’s wife Anna Ludwig dies
- 1923 Röntgen dies in Munich



Education

1862

Technical school in Utrecht

Barred from attending his finals due to disciplinary reasons

1864

Guest Lecturer at the University of Utrecht

1866

Federal Polytechnic institute, Zürich

Successfully admitted despite not graduating technical school

1868

University of Zürich

Completed his doctorate in studies on Gases

1870

University of Wurzburg

Assistant under his mentor, August Kundt. (he couldn't become a professor)

1874

University of Strasbourg

Finally able to become a professor

1876

Professor (various)

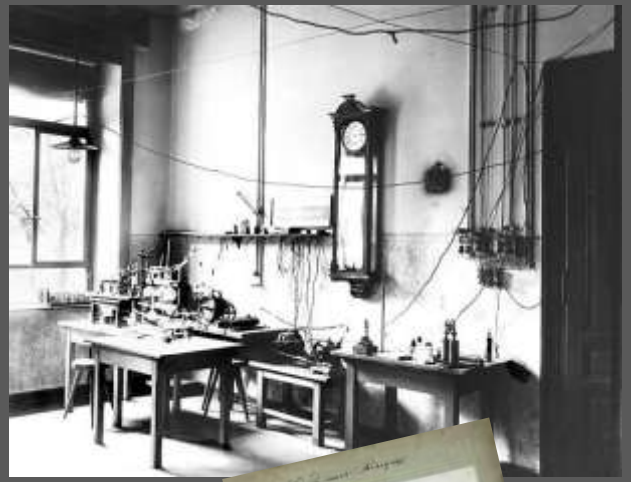
1876 - prof. of math & physics in Hohenheim
1879 - prof. at the University of Geissen
1888 - full prof. of physics at Wurzburg

1895

Discovery of X-rays

Discovery - X-ray

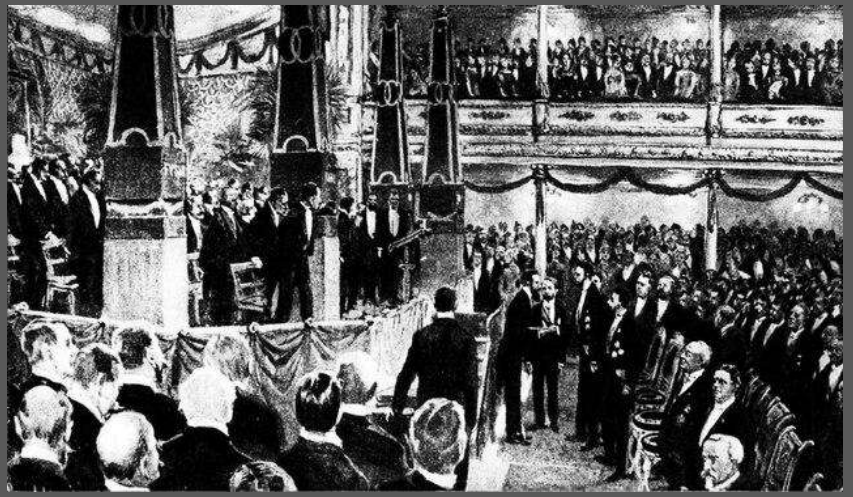
- Nov. 1895 when working with cathode tubes
  - After discovering that fluorescent beams escaped his dark cardboard shield, and hitting a nearby plate, causing it also to be fluorescent
  - By experimenting with different photographic and fluorescent materials he determined that this new ray had similar properties to light, but not being reflected or refracted.
  - This unknown (x) became the x-ray or Röntgenstrahlen
- Dec. 1895 First X-ray on human (his Wife),
  - “I have seen my death.” -Mrs. Röntgen





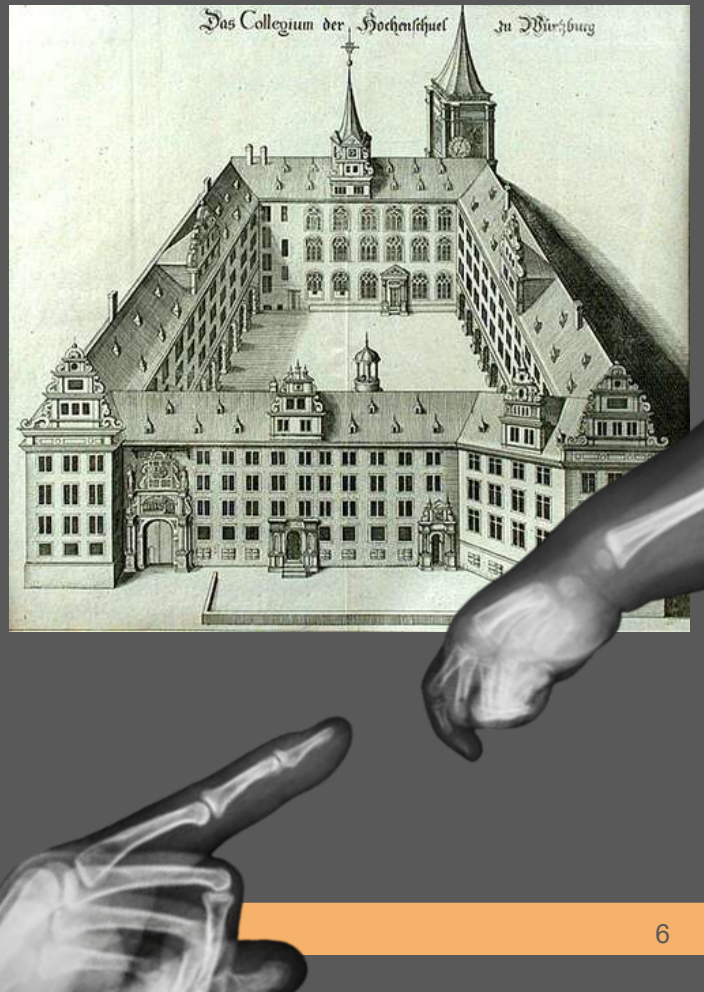
Achievements

- Won the first Nobel Prize in Physics for his discovery of X-Rays.
- Multiple international accolades, and recognition.
- Aided in the discovery of the electron (Knight).
- “Biggest” medical discovery after germ theory.



Faust in Röntgen - A

Röntgen embodies the Faustian Ideal of an endless pursuit of knowledge. He devoted his life to science and his studies. This is exemplified by his perseverance in becoming a professor and working on research, despite being set back by not being able to officially graduate from technical school.



Faust in Röntgen - B



In Röntgen's will he requested that specific records of his correspondences and experiments (even with x-rays) be destroyed after his death. This largely remains a mystery as to why this request was made, but in 2023 M. Treutwein made the following assertion, which shows the faustian ideal of knowledge for oneself (and its power):

> "A reason for Röntgen's decision might be found in his character: he had always wanted to publish only clear descriptions and evident results which had been thoroughly investigated. As such, he might have been afraid that after his death some papers and letters of more tentative nature could be published. Another reason might have been that Röntgen wanted to avoid a misuse against him" (Treutwein).

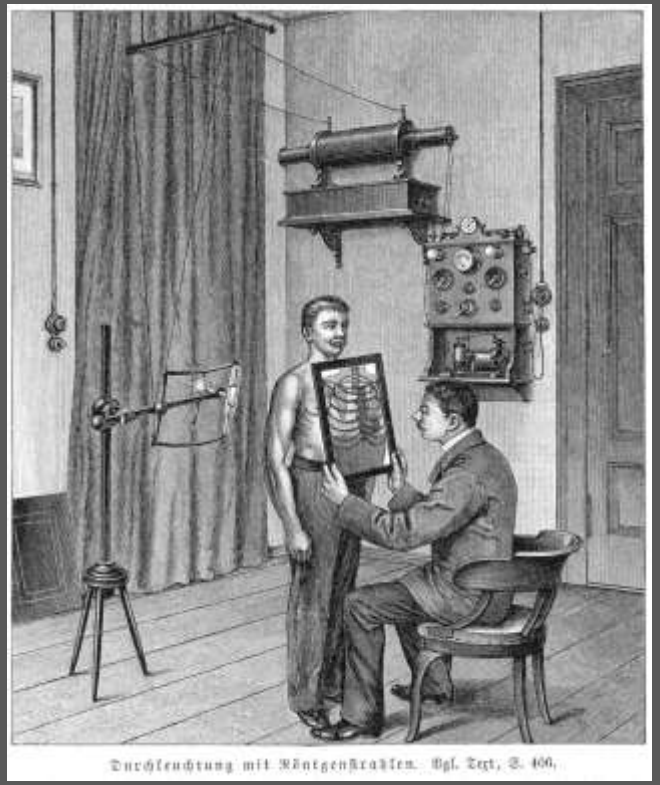
Faust in Röntgen - C

Röntgen’s pursuit of knowledge might have blinded him of the dangers of radiation, like many other scientists at the time, which in larger doses and prolonged exposure than the x-ray pose serious cancer risk. He might’ve also been unaware in the modern military applications of his invention. Displaying unintended consequences.



Faust (not) in Röntgen - D

Despite the examples shown before, Röntgen was described as selfless (exemplified by him adopting his brother's child), even refusing to patent his discoveries. He was truly a man of science contributing to the frontiers of physics and medicine.



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